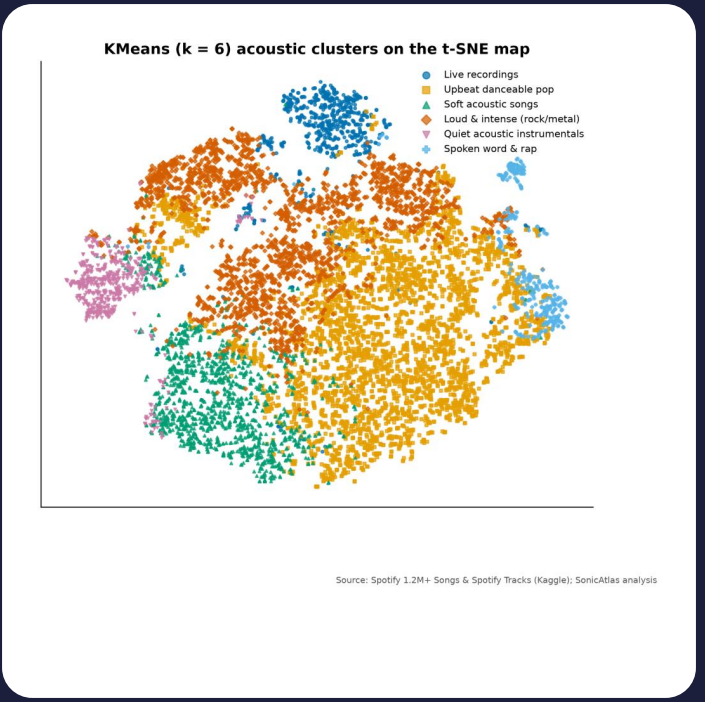
SONICATLAS

An Interactive In-Browser EDA Platform for Spotify Audio Features and Lyrics Across Genres and Eras

Review-3 | Implementation complete, evaluation and outcome

Team 05
25BDS0318 Sharvesh Prabhu
25BDS0308 Nithin S



Real acoustic map of 10,000 tracks (t-SNE)

BACSE301 Exploratory Data Analysis (Project-based Course)
Faculty: _____
School of Computer Science and Engineering, VIT Vellore | October 2026

Since Review-2: website, lyrics, network, tests and paper finished

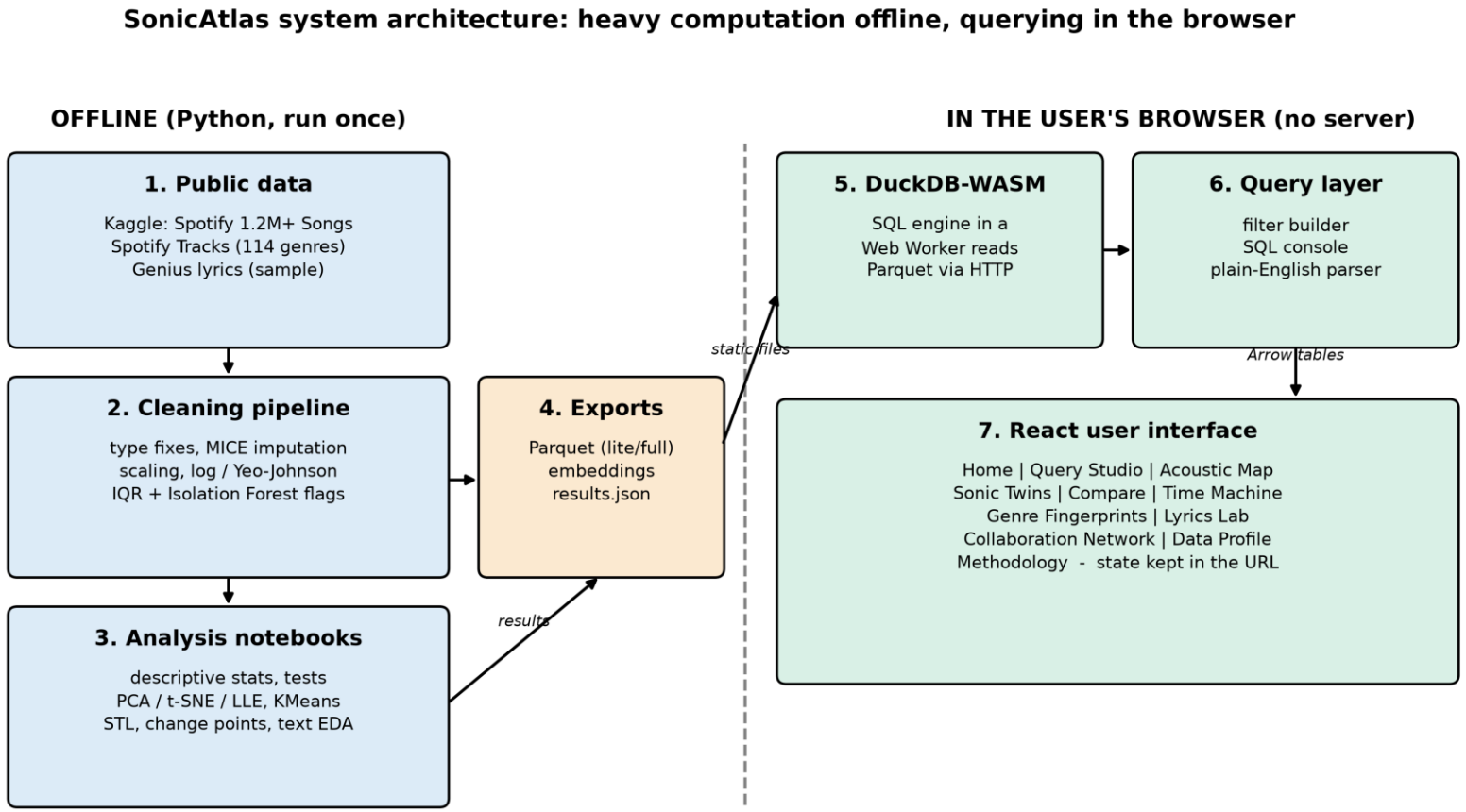
Problem and objectives

Open music data is messy and its insights are locked in static charts. Objectives: clean 1M+ tracks with validated repairs, apply the course EDA methods, compare embeddings, model trends, test genre differences, mine lyrics, and serve it all in a serverless website.

What changed since Review-2

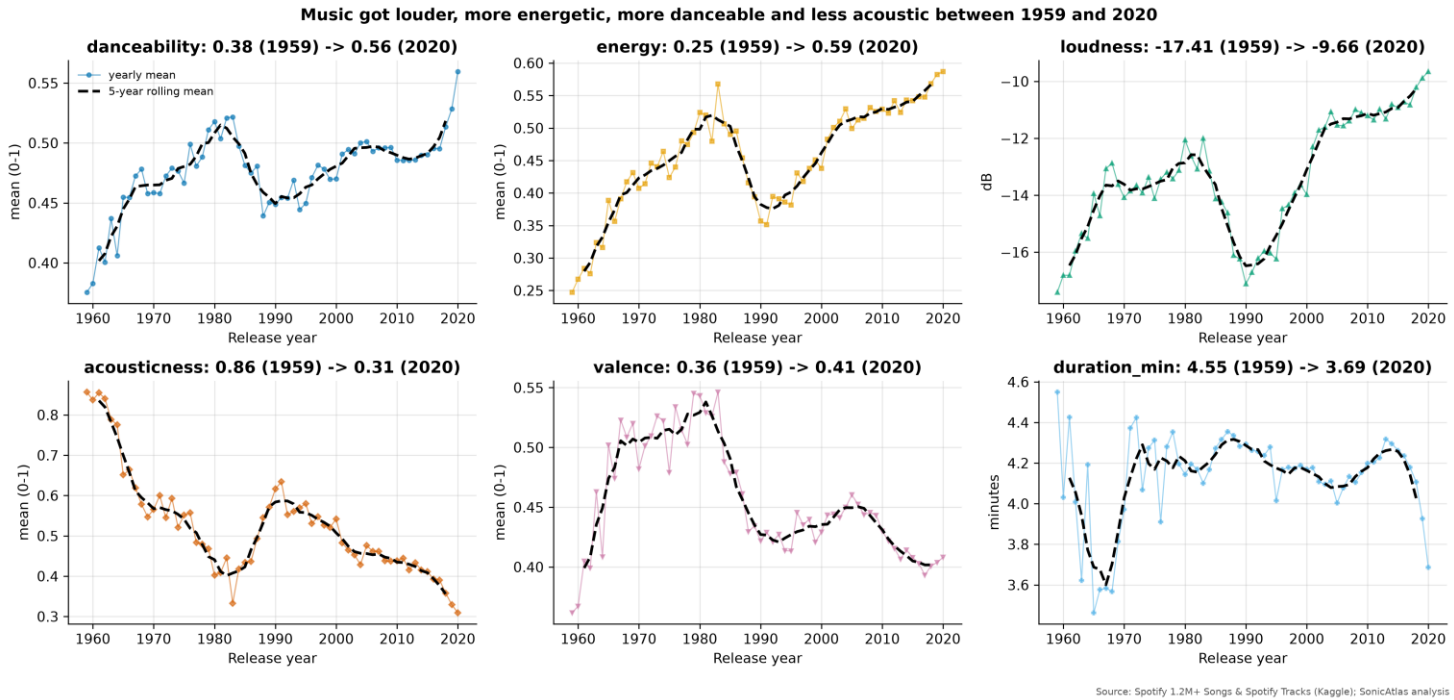
- Lyrics mining on 161,401 English songs
- Artist network: 84,132 artists, 189 communities
- Automated EDA benchmark (YData, Sweetviz, D-Tale)
- SonicAtlas website with 11 modules
- Tests, Lighthouse audit, IEEE paper, self-check

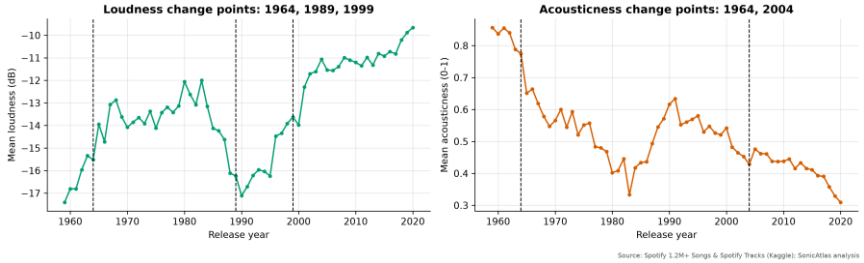
Heavy maths runs offline; querying runs in the user's browser



SonicAtlas design (Team 05)

Music got 7.8 dB louder and far less acoustic

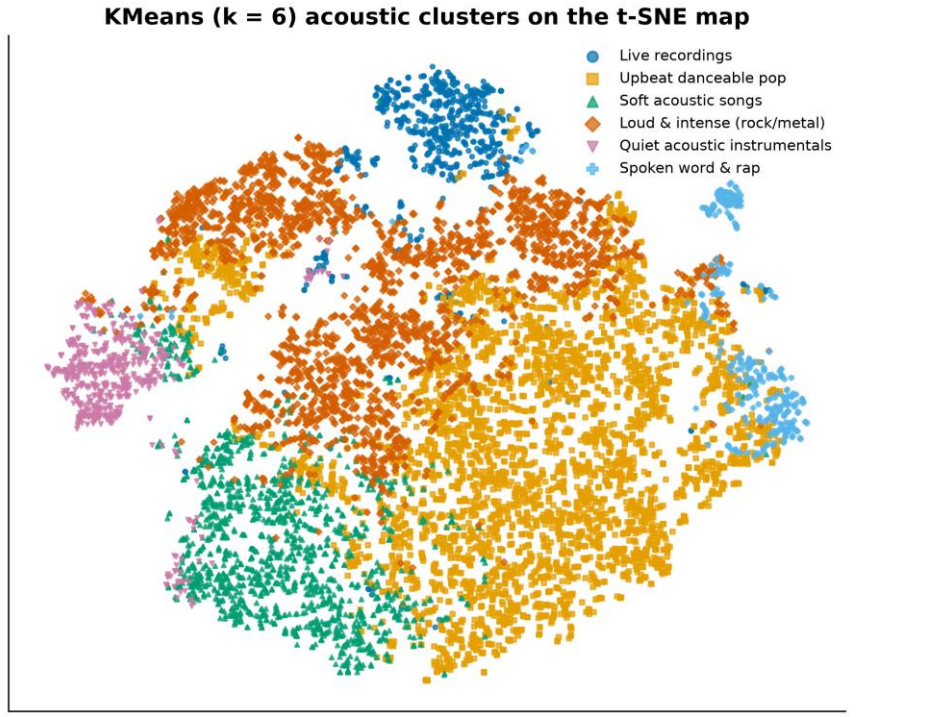




- Loudness -17.4 -> -9.7 dB (1959-2020)
- Acousticness 0.86 -> 0.31
- Loudness change points: 1989, 1999 ("loudness war")
- ARIMA MAE 0.58 dB vs naive 0.59 dB

Six named acoustic clusters, but genres overlap heavily

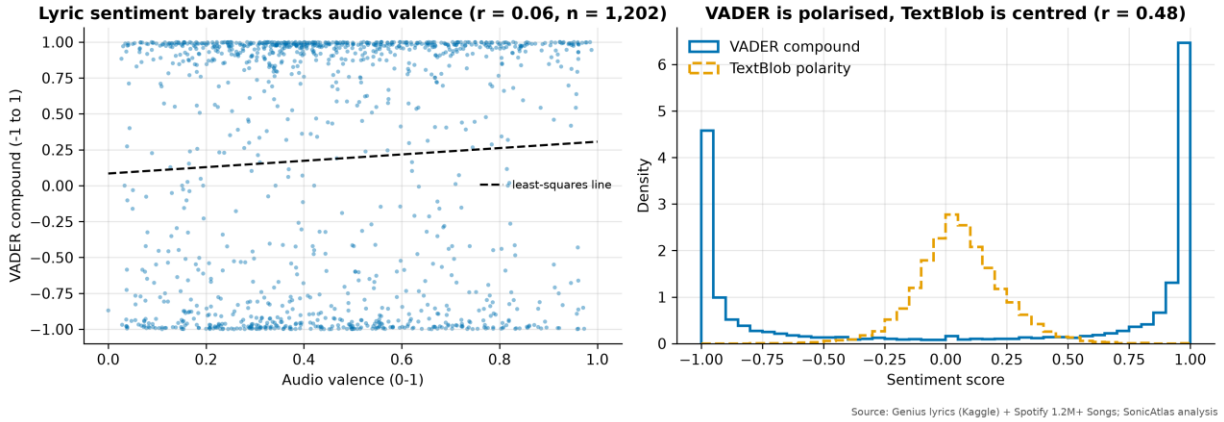
Cluster	Share	Top genres
Live recordings	6.2%	pagode, sertanejo, samba
Upbeat danceable pop	28.5%	salsa, dance, forro
Soft acoustic songs	20.5%	romance, tango, honky-tonk
Loud & intense (rock/metal)	21.9%	black-metal, grindcore, death-metal
Quiet acoustic instrumentals	18.7%	new-age, ambient, sleep
Spoken word & rap	4.2%	comedy, j-dance, dancehall



Source: Spotify 1.2M+ Songs & Spotify Tracks (Kaggle); SonicAtlas analysis

KMeans k = 6 on 7 PCs; HDBSCAN marks 65% as noise; genre silhouette is negative in every embedding.

Genre shapes danceability; lyric mood barely follows the music



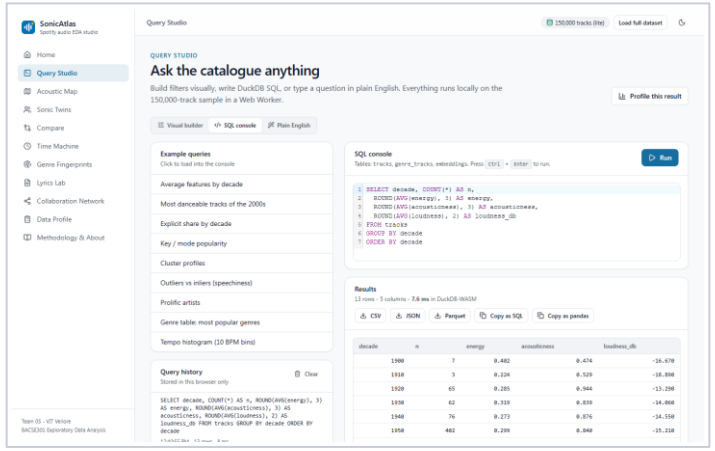
Test	Statistic	p	Effect size
ANOVA danceability x genre	F = 512.1	< 0.001	eta2 = 0.425
Kruskal-Wallis	H = 2061	< 0.001	eps2 = 0.423
Chi-square mode x genre	539.4	< 0.001	V = 0.333
Chi-square mode x decade	3089	< 0.001	V = 0.054

- 161,401 English songs; median 243 words, 24% repeated lines
- VADER vs TextBlob agree on sign for 71.0% (r = 0.48)
- Lyric sentiment vs audio valence r = 0.06 (n = 1,202)
- Country most positive, rap most negative

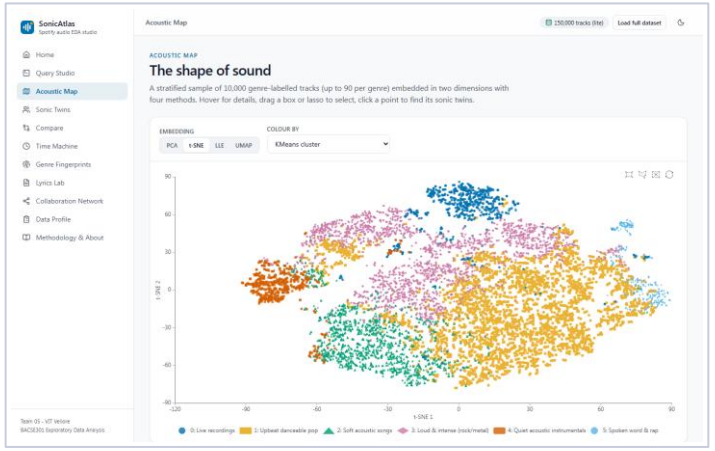
- Tukey HSD: 26/28 genre pairs differ
- Large n: read effect sizes, not p-values
- Top features: acousticness, loudness, energy

WEBSITE DEMO

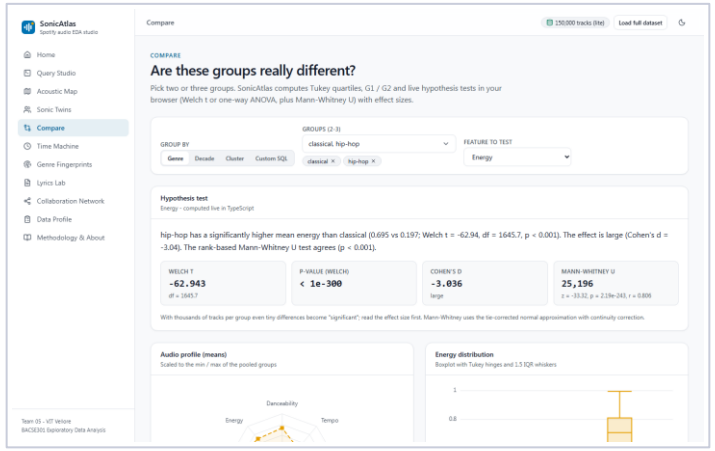
SonicAtlas runs every query inside the browser



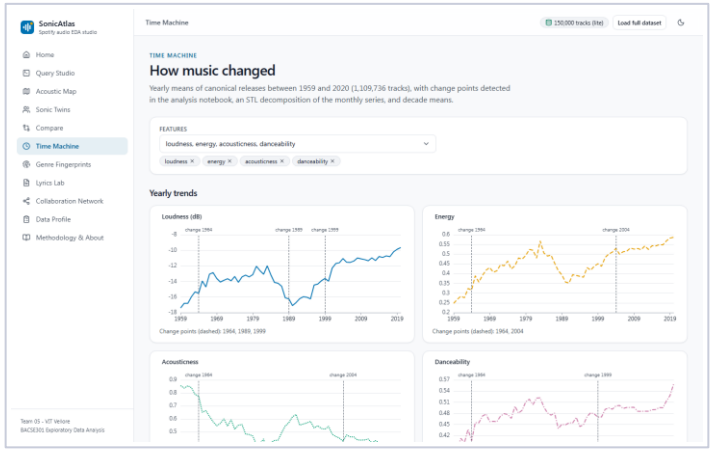
Query Studio



Acoustic Map



Compare



Time Machine

EVALUATION

Every component is measured, not just demonstrated

Method	Time (s)	Trust.	Silh.
PCA	0.0	0.779	-0.085
t-SNE	49.8	0.976	-0.061
LLE	42.4	0.671	-0.140
UMAP	32.2	0.947	-0.055

Automated EDA: YData 4.8 s, Sweetviz 6.3 s, D-Tale 4.1 s - fast overviews, but none flagged 0-BPM tempos or ran tests/embeddings.

4.7 ms

median lite-data query time

7/7

Python unit tests passed

65/65

web unit tests passed

8/8

Playwright end-to-end tests

98-100

Lighthouse performance, accessibility, best practice, SEO (4 pages)

OUTCOME

An IEEE paper, an open repository and a live tool



SonicAtlas: An Interactive In-Browser Exploratory Data Analysis Platform for Multi-Dimensional Audio Features and Lyrics of Spotify Tracks Across Genres and Eras

Sharvesh Prabhu (25BDS0318)
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
<email>

Nithin S (25BDS0308)
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
nithin.s2025a@vitstudent.ac.in

***Abstract*—Archived Spotify audio descriptors are among the few open windows onto the acoustic character of recorded music, but most analyses of them are static and stop at a handful of charts. We describe SonicAtlas, a reproducible exploratory pipeline and a serverless web platform built on 1,204,025 tracks, 89,740 genre-labelled tracks in 114 genres and a 161,401-song English lyrics sample. Impossible values are repaired with chained-equation imputation that lowers masked-value error by up to 57.5% relative to mean filling, and outliers are flagged rather than removed. Across 1959-2020, average loudness climbs by 7.8 dB while acousticness falls from 0.86 to 0.31, with loudness change points at 1989, 1999. Genre accounts for 42.5% of the variance of danceability, yet genre labels overlap strongly in every two-dimensional embedding; t-SNE attains the highest trustworthiness (0.976) among PCA, LLE, UMAP and t-SNE. Lyric sentiment is almost unrelated to audio valence (r = 0.06). All results are explorable in the browser through SQL executed by DuckDB-WASM on Parquet files, with no server.**

***Index Terms*—exploratory data analysis, music information retrieval, audio features, dimensionality reduction, time-series analysis, in-browser analytics**

I. Introduction

Popular music has been studied quantitatively for more than a decade, from the long-run homogenisation of timbre and loudness [1] to revolutions in US chart music [2] and the predictability of commercial success [3], [4]. Streaming platforms made such work easier by publishing per-track descriptors such as energy, valence and danceability, and large crawls of these values still circulate on open-data portals even though the original API endpoint has been closed to new applications.

Two practical problems limit the value of these crawls. First, the raw tables contain impossible values, duplicated re-issues and sources that do not share keys, and published studies rarely say how such issues were treated. Second, results are usually frozen into figures, so a reader cannot ask a slightly different question without rebuilding the pipeline. This paper addresses both problems. Our contributions are: (i) a documented cleaning pipeline over 1.2 million tracks with validated imputation and explicit outlier and re-issue flags; (ii) a systematic exploration that combines course-standard descriptive statistics, multivariate views, four embedding methods, clustering, time-series decomposition, hypothesis tests and lyrics mining; and (iii) SonicAtlas, a static web application in which every table can be filtered, queried and compared in the reader's own browser.

II. Related Work

Audio descriptors and genre. Large open collections such as the Million Song Dataset [5] and handcrafted features for genre recognition [6] established the use of numeric audio summaries, and deep models now learn such features directly from audio [7].

Trends and popularity. Studies of musical evolution [1], [2] and hit prediction [3], [4] report strong temporal signals but publish them as static results. Lyrics research shows rising negativity over time [8], usually without the matching audio view; sentiment is often scored with VADER [9]. Recommendation surveys [10] motivate similarity search over descriptors.

Embeddings and EDA tooling. PCA [11], LLE [12], t-SNE [13] and UMAP [14] are standard for maps of high-dimensional data. Automated EDA has been surveyed by Milo and Somech [15], and interactive charts are commonly specified with declarative grammars such as Vega-Lite [16]. On the systems side, embeddable analytical engines [17], their WebAssembly port [18] and scalable linked views [19] make serverless analysis feasible. The closest system in spirit is a Streamlit application for exploratory analysis of clinical data [20], which, unlike SonicAtlas, requires a Python server and is not domain-aware for music.

III. Datasets and Preprocessing

We combine three public Kaggle datasets: a crawl of 1,204,025 tracks with audio descriptors and release dates [21],

IEEE-format paper (6 pages) - status: prepared for submission

- IEEE CONECCT 2027: 29-31 July 2027, Bangalore, India
- ISMIR: ISMIR 2026 runs 8-12 November 2026 in Abu Dhabi
- IEEE VIS: VIS 2026 runs 9-13 November 2026 in Boston

- GitHub: [GitHub Link]
- Live site: [Live Website Link]
- IEEE checks: A4, 6 pages, fonts embedded yes, Type-3 fonts 0
- Similarity self-check: report 0.18%, paper 0.00% (not Turnitin)

Louder, less acoustic, and more than genre alone

Conclusion

Careful cleaning matters; music got louder and less acoustic; genre explains 42% of danceability but overlaps in every map; lyrics and sound carry different moods.

Limitations

Spotify features are model-derived; catalogue favours recent years; genre join only 10.7%; English-only lyrics; the 2020s have one year.

Future work

Open audio features, multilingual lyrics, user study.

[9] J. Serrà et al. (2012), Measuring the Evolution of Contemporary Western Popular Music
[11] C. Hutto et al. (2014), VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text
[16] L. van der Maaten et al. (2008), Visualizing Data using t-SNE
[25] D. H. da Silva et al. (2025), A Web Application for exploratory data analysis and classification of Parkinson's Disease patients using machine learning models on different datasets